

ASSIGNMENT

- a) What are vanishing and exploding gradients? How do they affect neural networks? Explain in detail.
- b) Use the Bank customer churn dataset from below Kaggle link and create an end-to-end project on Jupyter/Colab to predict the Churn.

<https://www.kaggle.com/datasets/santoshd3/bank-customers/data>

- i. Download the dataset from above link and load it into your Python environment.
 - ii. Perform the EDA and do the visualizations.
 - iii. Check the distributions/skewness in the variables and do the transformations if required.
 - iv. Check/Treat the outliers and do the feature scaling if required.
 - v. Build Deep Learning model using ANN with multiple hidden layers.
 - vi. Apply the dropout regularization and early stopping techniques to improve model performance.
 - vii. Use the modelCheckpoint also to store the parameters after each epoch.
 - viii. Use the KerasTuner to tune to best parameters (No. of hidden layers, optimizers, loss function, activation functions etc.)
 - ix. Compare the accuracies of different models and finalize the best model.
- c) Use the following Accident detection Dataset link and create an end-to-end project to predict Accident has happened or not.

<https://www.kaggle.com/datasets/ckay16/accident-detection-from-cctv-footage/data>

- i. Import the dataset and perform the EDA and do the visualizations.
- ii. Create model from scratch using your own number of filters, loss functions, no. of epochs and check the accuracy.
- iii. Now use the pretrained model to check if accuracy gets improved.
- iv. After that, go for data augmentation and further check if accuracy has improved.

- v. Compare the training and testing accuracy with all these 3 approaches and suggest the best model.
- d) Explain the role of padding and stride in convolutional layers? How will you decide when to use the padding.
- e) How does data augmentation help in improving CNN performance?